

1. Do user review scores on Steam accurately reflect video game quality?

No, Steam user review scores reflect player-perceived quality only moderately, with important genre, audience, and context biases.

Steam scores capture real player experience and often align with some quality dimensions, but they are not a clean or universal proxy for overall game quality

(Li 2021, Busurkina 2020). Evidence is strongest that reviews reveal perceived playability, satisfaction, and concrete flaws, while agreement with expert scores and broader quality constructs is inconsistent across genres and game types

(Park 2016, Fabricatore 2019).

Li, X., Zhang, Z. and Stefanidis, K. (2021). A Data-Driven Approach for Video Game Playability Analysis Based on Players' Reviews. *Inf.*, 12, pp. 129. doi: 10.3390/info12030129.

Busurkina, I., Karpenko, V., Tulubenskaya, E. and Bulygin, D. (2020). Game Experience Evaluation. A Study of Game Reviews on the Steam Platform., pp. 117-127. doi: 10.1007/978-3-030-65218-0_9.

Park, H.-S. and Byun, H.-W. (2016). Correlation Analysis: Game Professional Score and User Score on Steam. 11, pp. 237-246. doi: 10.14257/ijmue.2016.11.12.22.

Fabricatore, C., Gyaurov, D. and López, X. (2019). An exploratory study of the relationship between meaning-making and quality in games. *Multimedia Tools and Applications*, 78, pp. 13539 - 13564. doi: 10.1007/s11042-019-7232-1.

2. Do XGBoost, Random Forest, and gradient boosting ensemble methods outperform other classifiers on tabular binary classification benchmarks?

Yes, tree ensembles usually outperform other classifiers on tabular binary benchmarks, but not in every dataset.

Across broad tabular benchmarks, XGBoost, Random Forest, and related gradient boosting methods usually rank among the strongest classifiers, especially against linear models, SVMs, k-NN, and many deep tabular models (Shwartz-Ziv 2021, Uddin 2024, Grinsztajn 2022). The main caveat is that their advantage is not universal: some studies find small or statistically insignificant gaps among top tree ensembles, and some large or complex datasets favor newer deep models (Bentéjac 2019, Karlsson 2024).

Shwartz-Ziv, R. and Armon, A. (2021). Tabular Data: Deep Learning is Not All You Need. *Inf. Fusion*, 81, pp. 84-90. doi: 10.1016/j.inffus.2021.11.011.

Uddin, S. and Lu, H. (2024). Confirming the statistically significant superiority of tree-based machine learning algorithms over their counterparts for tabular data. *PLOS ONE*, 19. doi: 10.1371/journal.pone.0301541.

Grinsztajn, L., Oyallon, E. and Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data?. *Advances in Neural Information Processing Systems* 35. doi: 10.52202/068431-0037.

Bentéjac, C., Csörgo, A. and Martínez-Muñoz, G. (2019). A comparative analysis of gradient boosting algorithms. *Artificial Intelligence Review*, 54, pp. 1937 - 1967. doi: 10.1007/s10462-020-09896-5.

Karlsson, A., Wang, T., Nowaczyk, S., Pashami, S. and Asadi, S. (2024). Mind the Data, Measuring the Performance Gap Between Tree Ensembles and Deep Learning on Tabular Data., pp. 65-76. doi: 10.1007/978-3-031-58547-0_6.

3. How does BERT pre-training and Sentence-BERT sentence embeddings improve natural language understanding and downstream classification tasks?

BERT improves natural language understanding by learning bidirectional contextual representations from unlabeled text, then transferring all parameters into downstream models with minimal task-specific changes (Devlin 2019). Sentence-BERT improves sentence-level use of those representations by converting BERT into a siamese sentence encoder whose embeddings can be compared directly with cosine similarity (Reimers 2019).

Does BERT pre-training and Sentence-BERT improve natural language understanding and downstream classification tasks?

The literature strongly supports yes for both general language understanding and many downstream tasks, but the mechanism differs: BERT mainly improves token- and task-level contextualization, while SBERT mainly improves reusable sentence-level semantics and efficiency (Devlin 2019, Reimers 2019).

Pre-Training Effects

BERT's pre-training combines masked language modeling with next sentence prediction, letting it learn both word-level context and sentence-level relations before any task-specific supervision (Zhang 2023, Devlin 2019). Its input design also fuses token, segment, and position embeddings, which helps encode lexical identity, sentence-pair structure, and order information in a single representation (Zhang 2023, Devlin 2019).

Those pre-trained representations consistently improve downstream NLU because they resolve ambiguity and capture richer semantics than context-independent embeddings (Zhang 2023, Gao 2019). In the original BERT report, this transfer setup reached new state of the art on 11 NLP tasks, including GLUE, MultiNLI, and SQuAD, with large absolute gains over prior systems (Devlin 2019).

BERT's benefits also extend beyond vanilla text tasks when the same pre-training logic is adapted to richer structure. Segment-aware positional encoding in SegBERT outperformed original BERT on various NLP tasks, suggesting that richer document structure can further improve contextual representations (Bai 2020). Multimodal extensions such as VL-BERT and Pixel-BERT show that pre-training aligns language with visual signals and improves downstream visual reasoning, retrieval, and VQA tasks (Su 2019, Huang 2020).

Devlin, J., Chang, M.-W., Lee, K. and Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding., pp. 4171-4186. doi: 10.18653/v1/n19-1423.

Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. ArXiv, abs/1908.10084. doi: 10.18653/v1/d19-1410.

Zhang, X., Wu, Z., Liu, K., Zhao, Z., Wang, J. and Wu, C. (2023). Text Sentiment Classification Based on BERT Embedding and Sliced Multi-Head Self-Attention Bi-GRU. Sensors (Basel, Switzerland), 23. doi: 10.3390/s23031481.

Gao, Z., Feng, A., Song, X. and Wu, X. (2019). Target-Dependent Sentiment Classification With BERT. IEEE Access, 7, pp. 154290-154299. doi: 10.1109/access.2019.2946594.

Bai, R. H., Shi, P., Lin, J. J., Tan, L., Xiong, K., Gao, W. and Li, M. (2020). SegBERT: Pre-training of Segment-aware BERT for Language Understanding. ArXiv, abs/2004.14996.

Su, W., Zhu, X., Cao, Y., Li, B., Lu, L., Wei, F. and Dai, J. (2019). VL-BERT: Pre-training of Generic Visual-Linguistic Representations. ArXiv, abs/1908.08530.

Huang, Z., Zeng, Z., Liu, B., Fu, D. and Fu, J. (2020). Pixel-BERT: Aligning Image Pixels with Text by Deep Multi-Modal Transformers. ArXiv, abs/2004.00849.

4. How does LIME local interpretable model-agnostic explanation provide post-hoc interpretability for black-box machine learning classifiers?

LIME provides post-hoc interpretability by fitting a simple local surrogate around one black-box prediction.

LIME gives post-hoc interpretability to a black-box classifier by explaining one prediction at a time with an interpretable model trained after the original model is already built

(Ribeiro 2016 ,Yasui 2025). It is model-agnostic because the explainer only needs the black box's input-output behavior near the instance, not access to its internals

(Ribeiro 2016, Raza2025).

Core Mechanism

LIME's goal is to learn an interpretable model in an interpretable representation that is locally faithful to the classifier near the instance being explained

(Ribeiro 2016). In practice, it explains the influence of each feature on the outcome for a single subject or case, often showing class probabilities and per-feature contributions visually(Salih 2023).

It does this by generating perturbed or simulated samples around the target instance, querying the black-box classifier on those nearby samples, and then fitting a simple surrogate such as a sparse linear model, linear classifier, decision tree, or linear regression model to mimic the local behavior

(Zafar 2021 ,Zafar 2019 Yasui 2025). Samples closer to the target instance are emphasized, so the surrogate reflects the classifier's behavior in the immediate neighborhood rather than globally (Yasui 2025, An 2023).

Ribeiro, M. T., Singh, S. and Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. doi: 10.1145/2939672.2939778.

Yasui, D. and Sato, H. (2025). Improving Local Fidelity and Interpretability of LIME by Replacing Only the Sampling Process With CVAE. *IEEE Access*, 13, pp. 53084-53099. doi: 10.1109/access.2025.3553505.

Raza, R., Wang, G., Wong, K. W., Laga, H. and Fisichella, M. (2025). ITL-LIME: Instance-Based Transfer Learning for Enhancing Local Explanations in Low-Resource Data Settings. *Proceedings of the 34th ACM International Conference on Information and Knowledge Management*. doi: 10.1145/3746252.3761183.

Salih, A. M. A., Raisi-Estabragh, Z., Galazzo, I., Radeva, P., Petersen, S. E., Lekadir, K. and Menegaz, G. (2023). A Perspective on Explainable Artificial Intelligence Methods: SHAP and LIME. *Advanced Intelligent Systems*, 7. doi: 10.1002/aisy.202400304.

Zafar, M. R. and Khan, N. (2021). Deterministic Local Interpretable Model-Agnostic Explanations for Stable Explainability. *Mach. Learn. Knowl. Extr.*, 3, pp. 525-541. doi: 10.3390/make3030027.

Zafar, M. R. and Khan, N. (2019). DLIME: A Deterministic Local Interpretable Model-Agnostic Explanations Approach for Computer-Aided Diagnosis Systems. *ArXiv*, abs/1906.10263.

An, J., Zhang, Y. and Joe, I. (2023). Specific-Input LIME Explanations for Tabular Data Based on Deep Learning Models. *Applied Sciences*. doi: 10.3390/app13158782.

5. What methods effectively address class imbalance in binary classification, and how does SMOTE oversampling compare to class weighting and undersampling strategies?

Overall, effective methods for binary class imbalance are resampling, class weighting, threshold tuning, and ensemble hybrids, with no single approach dominating every dataset. SMOTE is widely useful and often better than no treatment, but class weighting is

competitive and undersampling or hybrid methods often win when data are noisy, overlapping, or high-dimensional (Abdelhamid 2024, Carvalho 2025, Blagus 2013).

Abdelhamid, M. and Desai, A. (2024). Balancing the Scales: A Comprehensive Study on Tackling Class Imbalance in Binary Classification. ArXiv, abs/2409.19751. doi: 10.48550/arxiv.2409.19751.

Carvalho, M., Pinho, A. J. and Brás, S. (2025). Resampling approaches to handle class imbalance: a review from a data perspective. Journal of Big Data, 12. doi: 10.1186/s40537-025-01119-4.

Blagus, R. and Lusa, L. (2013). SMOTE for high-dimensional class-imbalanced data. BMC Bioinformatics, 14, pp. 106 - 106. doi: 10.1186/1471-2105-14-106.

6. What is the economic scale of the global video game industry and how do user review scores affect game commercial success and developer sustainability?

Global Gaming Market and Review-Driven Success

The global video game industry is a very large, still-growing entertainment sector, with reported global market estimates ranging from about \$152 billion to \$300 billion depending on year and what revenue streams are counted, and review signals appear to meaningfully affect sales, though sustainability evidence for developers is more indirect and mixed

(Ilyanov 2020, Peri 2024, Fan2023).

The literature supports a yes on commercial success: higher user and critic review valence tends to predict higher sales or downloads, but the size of that effect depends on platform, review type, and game segment

(Kraemer 2025, Sällberg 2022, Adiguzel 2021).

Ilyanov, D. S., Chernysheva, T. C. and Yurevich, M. (2020). Sources of economic growth in the XXI century: video game industry. Теоретическая и прикладная экономика. doi: 10.25136/2409-8647.2020.3.31693.

Peri, J. (2024). Video gaming industry in the US. International Journal of Science and Research Archive. doi: 10.30574/ijrsra.2024.11.1.0194.

Fan, N. (2023). Evaluation of the Performance and Behavior of the Video Game Industry. Advances in Economics, Management and Political Sciences. doi: 10.54254/2754-1169/30/20231425.

Kraemer, T., Weiger, W. H. and Heidenreich, S. (2025). Do all stars shine the same? Investigating the nonlinear effects of user and critic reviews on video game sales. Journal of Business Research. doi: 10.1016/j.jbusres.2024.115034.

Sällberg, H., Wang, S. and Numminen, E. (2022). The combinatory role of online ratings and reviews in mobile app downloads: an empirical investigation of gaming and productivity apps from their initial app store launch. Journal of Marketing Analytics, 11, pp. 426-442. doi: 10.1057/s41270-022-00171-w.

Adiguzel, F. (2021). The Effect of YouTube Reviews on Video Game Sales. Journal of Business Research-Turk, 13, pp. 2096-2109. doi: 10.20491/isarder.2021.1249.

7. Can machine learning predict commercial success or audience reception using only pre-launch or pre-release metadata, and what methods are most effective across different product domains?

Overall Evidence

Across films, fashion, games, music, and consumer products, studies repeatedly report that pre-release prediction is feasible using only information available before launch

(Souza 2023, Lee 2014, Arampatzis 2023). In film, this includes genre, cast, director, plot, budget, and related metadata (Krishnan 2025, Memon 2023, Fahoum 2023).

Evidence is strongest that adding prerelease demand signals beats plain metadata models. Online community awareness, word-of-mouth, expectations, and adoption intention increased opening-week movie sales prediction beyond generic variables (Divakaran 2017). Prerelease Google Trends improved video-game forecasts by more than 18% in one study and up to 21% for life-cycle sales in another (Schaer 2022, Schaer 2025).

Souza, T. L. (2023). Machine Learning for Multimedia Movie Success Prediction.. doi: 10.11606/t.100.2023.tde-16012024-220439.

Lee, H., Kim, S.-G., Park, H.-W. and Kang, P. (2014). Pre-launch new product demand forecasting using the Bass model: : A statistical and machine learning-based approach. Technological Forecasting and Social Change, 86, pp. 49-64. doi: 10.1016/j.techfore.2013.08.020.

Arampatzis, M., Theodoridis, G. and Tsadiras, A. (2023). Pre-launch Fashion Product Demand Forecasting Using Machine Learning Algorithms., pp. 362-372. doi: 10.1007/978-3-031-34107-6_29.

Krishnan, H., Sankar, J., Baby, M. K., Ali, N. and Ramesh, S. (2025). Movie Success Forecasting and Audience Clustering Using XGBoost and Fuzzy C-Means. 2025 Advanced Computing and Communication Technologies for High Performance Applications (ACCTHPA), pp. 1-6. doi: 10.1109/accthpa65749.2025.11168537.

Memon, Z. and Hussain, S. M. (2023). Predicting movie success based on pre-released features. Multimedia Tools and Applications, 83, pp. 20975 - 20996. doi: 10.1007/s11042-023-16319-4.

Fahoum, A. A. S. and Ghobon, T. A. (2023). Performance Predictions of Sci-Fi Films via Machine Learning. *Applied Sciences*. doi: 10.3390/app13074312.

Divakaran, P. K. P., Palmer, A., Søndergaard, H. A. and Matkovskyy, R. (2017). Pre-launch Prediction of Market Performance for Short Lifecycle Products Using Online Community Data. *Journal of Interactive Marketing*, 38, pp. 12 - 28. doi: 10.1016/j.intmar.2016.10.004.

Schaer, O., Kourentzes, N. and Fildes, R. (2022). Predictive competitive intelligence with prerelease online search traffic. *Production and Operations Management*, 31, pp. 3823 - 3839. doi: 10.1111/poms.13790.

Schaer, O., Kourentzes, N. and Fildes, R. (2025). Analogy-based life cycle forecasts with pre-release buzz. *European Journal of Operational Research*. doi: 10.1016/j.ejor.2025.12.003.

8. How do stacking ensemble methods and meta-learner strategies improve multi-model classification performance, and how can prediction confidence be used for adaptive model selection?

Stacking and confidence-based selection answer two linked sub-questions: how to combine diverse models more effectively, how to choose among them adaptively for each sample, and what limits those strategies in practice. Across domains, stacking usually beats single models and often other ensemble schemes, but gains depend on base-model diversity, meta-learner choice, and calibration quality (Müller 2022, Nti 2020, Kablan 2023).

Müller, D., Soto-Rey, I. and Kramer, F. (2022). An Analysis on Ensemble Learning Optimized Medical Image Classification With Deep Convolutional Neural Networks. *IEEE Access*, 10, pp. 66467-66480. doi: 10.1109/access.2022.3182399.

Nti, I. K., Adekoya, A. F. and Weyori, B. (2020). A comprehensive evaluation of ensemble learning for stock-market prediction. *Journal of Big Data*, 7. doi: 10.1186/s40537-020-00299-5.

Kablan, R., Miller, H. A., Suliman, S. and Frieboes, H. (2023). Evaluation of stacked ensemble model performance to predict clinical outcomes: A COVID-19 study. *International Journal of Medical Informatics*, 175, pp. 105090 - 105090. doi: 10.1016/j.ijmedinf.2023.105090.

9. How do hybrid models combining pre-trained language model embeddings with structured tabular features compare to tabular-only or text-only approaches in binary classification?

Hybrid models that combine pre-trained language model embeddings with structured features usually beat text-only baselines, but they do not consistently beat strong tabular-only models such as gradient-boosted trees. Gains are largest when tables contain semantically meaningful text fields, when modalities interact directly, or when fusion is more structured than simple concatenation (Liu 2022, Liu 2022).

Liu, G., Yang, J. and Wu, L. Y. (2022). PTab: Using the Pre-trained Language Model for Modeling Tabular Data. *ArXiv*, abs/2209.08060. doi: 10.48550/arxiv.2209.08060.

Liu, S., Wang, X., Hou, Y., Li, G., Wang, H., Xu, H., Xiang, Y. and Tang, B. (2022). Multimodal Data Matters: Language Model Pre-Training Over Structured and Unstructured Electronic Health Records. *IEEE Journal of Biomedical and Health Informatics*, 27, pp. 504-514. doi: 10.1109/jbhi.2022.3217810.

10. How reliable are user review scores as quality signals for video games, and what factors drive discrepancies between user and critic game ratings?

User review scores for video games appear informative but imperfect: they predict demand and capture player experience, but they are more polarized, more vulnerable to manipulation, and less stable than critic scores (Kraemer 2025, Santos2019, Coronado-BI'azquez 2024)

. The main sub-questions are signal reliability, sources of critic-user divergence, and when disagreement matters for decisions

(Santos 2019, Wan2024).

Kraemer, T., Weiger, W. H. and Heidenreich, S. (2025). Do all stars shine the same? Investigating the nonlinear effects of user and critic reviews on video game sales. *Journal of Business Research*. doi: 10.1016/j.jbusres.2024.115034.

Santos, T., Lemmerich, F., Strohmaier, M. and Helic, D. (2019). What's in a Review. *Proceedings of the ACM on Human-Computer Interaction*, 3, pp. 1 - 22. doi: 10.1145/3359242.

Coronado-BI'azquez, J. (2024). A NLP Approach to "Review Bombing" in Metacritic PC Videogames User Ratings. *ArXiv*, abs/2405.06306. doi: 10.48550/arxiv.2405.06306.

Wan, M. S. K. and Sam, A. F. (2024). From praise to polarization: the role of controversial reviews in shaping consumer behavior in the PC game market. *Journal of Marketing Analytics*. doi: 10.1057/s41270-024-00359-2.

11. Which evaluation metrics are most appropriate for imbalanced binary classification problems, and why is accuracy misleading for skewed class distributions?

For imbalanced binary classification, the most appropriate metrics depend on the decision goal, but the literature most consistently favors balanced accuracy for overall error across classes, Matthews correlation coefficient (MCC) when all four confusion-matrix cells should matter, and precision/recall-based metrics when the minority positive class is the main target (Thölke 2023, Chicco 2020, Saito 2015). Accuracy is misleading because it

weights correct predictions by class prevalence, so a classifier can score very high by predicting the majority class and still fail badly on the minority class (Thölke 2023, Hedao2024).

Thölke, P., Mantilla-Ramos, Y.-J., Abdelhedi, H., Maschke, C., Dehgan, A., Harel, Y., Kemtur, A., Berrada, L. M., Sahraoui, M., Young, T., Pépin, A. B., Khantour, C. E., Landry, M., Pascarella, A., Hadid, V., Combrisson, E., O’Byrne, J. and Jerbi, K. (2023). Class imbalance should not throw you off balance: Choosing the right classifiers and performance metrics for brain decoding with imbalanced data. *bioRxiv*. doi: 10.1101/2022.07.18.500262.

Chicco, D. and Jurman, G. (2020). The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics*, 21. doi: 10.1186/s12864-019-6413-7.

Saito, T. and Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLoS ONE*, 10. doi: 10.1371/journal.pone.0118432.

Hedao, A., Gaikwad, V., Ghadekar, P. and Jalnekar, R. (2024). Choosing the best metrics for quantifying the quality of the model in skewed binary classification problems. *AIP Conference Proceedings*. doi: 10.1063/5.0177799.

12. How does feature engineering, interaction terms, and derived feature creation improve supervised machine learning performance on structured datasets?

Feature engineering improves supervised learning on structured datasets through three linked mechanisms: better representations, explicit interactions, and task-specific derived variables (Tschalzev 2024, Verdonck 2021). Across the literature, gains are strongest when raw columns do not already align with the model’s inductive bias, especially for categorical data, sparse settings, and domains where expert or automated transforms reveal relationships the base model would not reliably discover (Tschalzev 2024, Azim 2023).

Tschalzev, A., Marton, S., Ludtke, S., Bartelt, C. and Stuckenschmidt, H. (2024). A Data-Centric Perspective on Evaluating Machine Learning Models for Tabular Data. ArXiv, abs/2407.02112. doi: 10.48550/arxiv.2407.02112.

Verdonck, T., Baesens, B., Óskarsdóttir, M. and Broucke, S. V. (2021). Special issue on feature engineering editorial. Machine Learning, 113, pp. 3917 - 3928. doi: 10.1007/s10994-021-06042-2.

Azim, E., Wang, D., Liu, K., Zhang, W. and Fu, Y. (2023). Feature Interaction Aware Automated Data Representation Transformation., pp. 878-886. doi: 10.1137/1.9781611978032.100.